01:960:490 - Homework 1

Pranav Tikkawar

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1. 2.21

   A normally distributed random variable has an unknown mean μ and a known variance $\sigma^2 = 9$. Find the sample size required to construct a 95% confidence interval for the mean that has a total length of 1.0.

Solution:

```
> ## 2.21: sample size for CI length
>
> sigma2 <- 9          # given variance
> sigma  <- sqrt(sigma2)
> L      <- 1.0        # desired TOTAL length of CI
> conf   <- 0.95
>
> alpha  <- 1 - conf
> z      <- qnorm(1 - alpha/2)        # z_{0.975} for 95% CI
>
> # Formula: L = 2 * z * sigma / sqrt(n)  =>  n = (2*z*sigma / L)^2
> n_exact <- (2 * z * sigma / L)^2
> n_exact
[1] 138.2925
>
> # Required integer sample size (round up)
> n_required <- ceiling(n_exact)
> n_required
[1] 139
```

Using the two-sided normal-theory confidence interval for a mean with known variance, the total length of the interval is

$$L = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

We are given a desired total length $L = 1$, a variance $\sigma^2 = 9$ (so $\sigma = 3$), and a 95% confidence level, which implies

$$\alpha = 0.05 \quad \text{and} \quad z_{\alpha/2} = z_{0.025} \approx 1.96.$$

Substituting into the length formula and solving for n yields

$$1 = 2 \cdot 1.96 \cdot \frac{3}{\sqrt{n}} \quad \Rightarrow \quad \sqrt{n} = 2 \cdot 1.96 \cdot 3 \approx 11.76 \quad \Rightarrow \quad n \approx 138.3.$$

Because the sample size must be an integer and we require the confidence interval length to be *at most* 1.0, we round up to obtain

$$n = 139.$$

Thus, a sample size of 139 observations is required.

2. 2.22

   The shelf life of a carbonated beverage is of interest. Ten bottles are randomly selected and tested, and the following results (in days) are obtained:

   $$108,\ 124,\ 124,\ 106,\ 115,\ 138,\ 163,\ 159,\ 134,\ 139.$$

   (a) We would like to demonstrate that the mean shelf life exceeds 120 days. Set up appropriate hypotheses for investigating this claim.

   (b) Test these hypotheses using $\alpha = 0.01$. What are your conclusions?

   (c) Find the P-value for the test in part (b).

   (d) Construct a 99 percent confidence interval on the mean shelf life.

Solution:

```
> # 2.22
> ## Data: shelf life in days
> x <- c(108, 124, 124, 106, 115, 138, 163, 159, 134, 139)
>
> ## Basic summaries
> n  <- length(x)
> xbar <- mean(x)
> s  <- sd(x)
> n; xbar; s
[1] 10
[1] 131
[1] 19.54482
>
> ## (b) One-sample t test: H0: mu = 120 vs Ha: mu > 120, alpha = 0.01
> mu0 <- 120
> # Test statistic (by hand)
> t_stat <- (xbar - mu0) / (s / sqrt(n))
> df <- n - 1
> t_stat
[1] 1.779758
> df
[1] 9
> # Critical value for alpha = 0.01, right-tailed
> t_crit <- qt(0.99, df = df)   # t_{0.99,9}
> t_crit
[1] 2.821438
>
> ## (c) P-value for right-tailed test
> p_value <- 1 - pt(t_stat, df = df)  # P(T >= t_stat)
> p_value
[1] 0.05440887
>
> ## (d) 99% confidence interval for the mean (by hand)
> alpha <- 0.01
> t_crit_2sided <- qt(1 - alpha/2, df = df)  # t_{0.995,9}
> SE <- s / sqrt(n)
> ME <- t_crit_2sided * SE
> lower_99 <- xbar - ME
> upper_99 <- xbar + ME
> c(lower_99, upper_99)
[1] 110.914 151.086
```

For this problem the data (in days) are

$$108,\ 124,\ 124,\ 106,\ 115,\ 138,\ 163,\ 159,\ 134,\ 139,$$

giving

$$n = 10, \qquad \bar{x} = 131, \qquad s = 19.54482.$$

(a) We want to show the mean shelf life exceeds 120 days, so we test

$$H_0:\ \mu = 120 \qquad \text{vs} \qquad H_a:\ \mu > 120.$$

(b) Using a one-sample t-test with unknown variance, the test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{131 - 120}{19.54482/\sqrt{10}} = 1.779758,$$

with $df = n - 1 = 9$. For a right-tailed test at level $\alpha = 0.01$, the critical value is

$$t_{0.99,9} = 2.821438.$$

Since $1.779758 < 2.821438$, we do not reject H_0 at $\alpha = 0.01$; there is not enough evidence at the 1% level to conclude that the mean shelf life exceeds 120 days.

(c) The P-value for the right-tailed test is

$$P\text{-value} = P(T_9 \geq 1.779758) = 0.05440887.$$

Because $0.0544 > 0.01$, this again leads us to fail to reject H_0 at the 1% significance level.

(d) A 99% confidence interval for the mean is

$$\bar{x} \pm t_{0.995,9}\,\frac{s}{\sqrt{n}},$$

where $t_{0.995,9} = 3.249836$. With

$$SE = \frac{s}{\sqrt{n}} = \frac{19.54482}{\sqrt{10}}, \qquad ME = t_{0.995,9}\,SE,$$

we obtain

$$(\bar{x} - ME,\ \bar{x} + ME) = (110.914,\ 151.086).$$

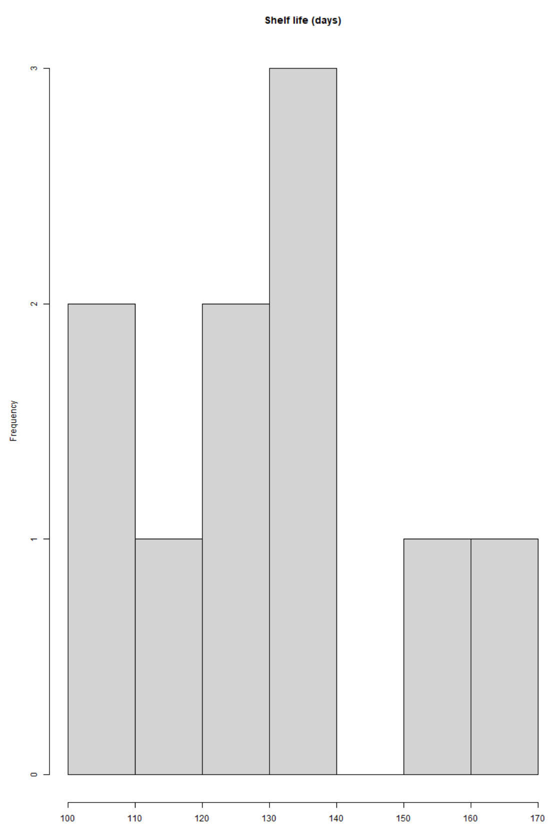
Thus, a 99% confidence interval for μ is approximately $(110.914,\ 151.086)$ days.

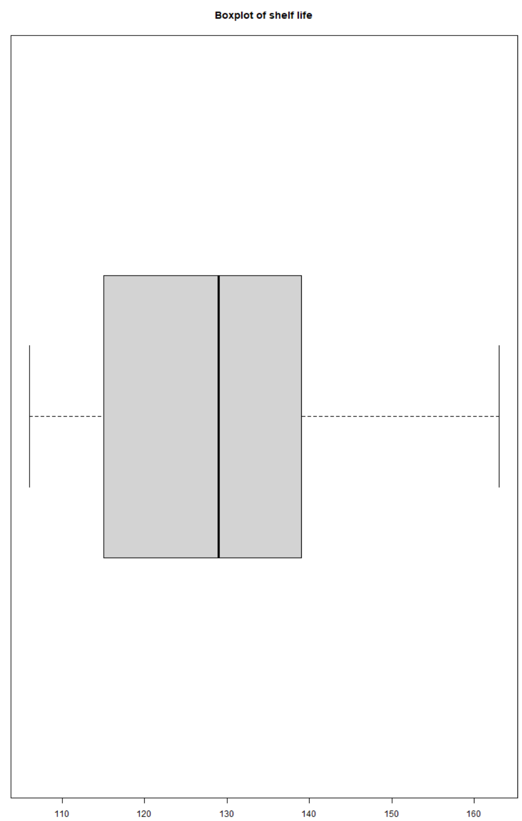
3. 2.23

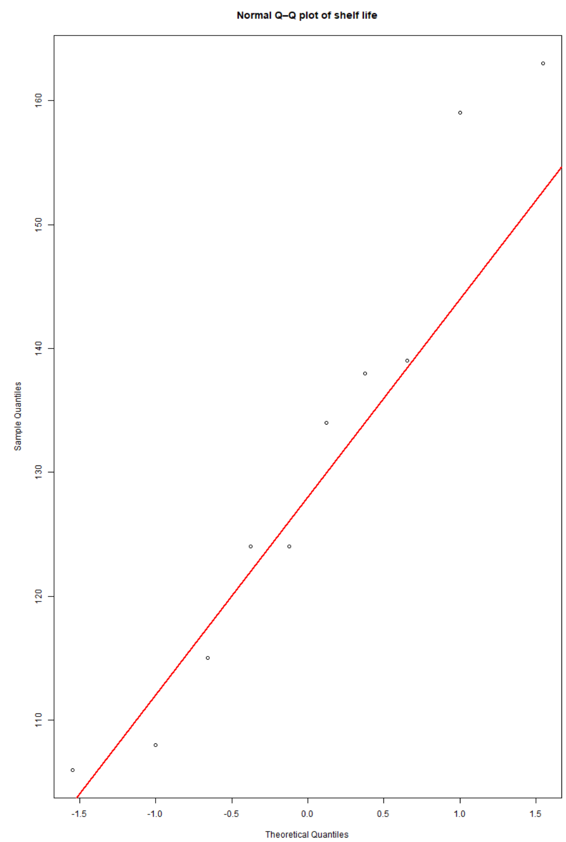
Consider the shelf life data in Problem 2.22. Can shelf life be described or modeled adequately by a normal distribution? What effect would the violation of this assumption have on the test procedure you used in solving Problem 2.17?

Solution:

```
> ## 2.23  Normality check for shelf life data (from 2.22)
>
> x <- c(108, 124, 124, 106, 115, 138, 163, 159, 134, 139)
> ## Basic histogram and boxplot
> hist(x,
+      main = "Shelf life (days)",
+      xlab = "Days")
> boxplot(x,
+         horizontal = TRUE,
+         main = "Boxplot of shelf life")
> ## Normal Q-Q plot
> qqnorm(x,
+        main = "Normal Q-Q plot of shelf life")
> qqline(x, col = "red", lwd = 2)
> ## Shapiro-Wilk normality test
> shapiro.test(x)

        Shapiro-Wilk normality test

data:  x
W = 0.93654, p-value = 0.5152
```




Boxplot of shelf life
110
120
130
140
150
160



Using the data from Problem 2.22, a histogram, boxplot, and normal Q–Q plot show no extreme outliers and only mild deviation from a straight line. The Shapiro–Wilk test also does not provide strong evidence against normality. Therefore, it is reasonable to model the shelf life data with a normal distribution.

The test in Problem 2.22 was a one-sample t-test for the mean. This procedure is fairly robust to mild departures from normality, especially when there are no strong outliers. Thus, small violations of normality would have little practical effect on the validity of the test and its P-value or confidence interval. However, if the true distribution were highly skewed or heavy-tailed, the actual significance level could differ from the nominal value, and the conclusions from the t-test might be less reliable.

4. 2.31

Photoresist is a light-sensitive material applied to semiconductor wafers so that the circuit pattern can be imaged on to the wafer. After application, the coated wafers are baked to remove the solvent in the photoresist mixture and to harden the resist. Here are measurements of photoresist thickness (in kA) for eight wafers baked at two different temperatures. Assume that all of the runs were made in random order.

95°C	100°C
11.176	5.263
7.089	6.748
8.097	7.461
11.739	7.015
11.291	8.133
10.759	7.418
6.467	3.772
8.315	8.963

(a) Is there evidence to support the claim that the higher baking temperature results in wafers with a lower mean photoresist thickness? Use $\alpha = 0.05$.

(b) What is the P-value for the test conducted in part (a)?

(c) Find a 95 percent confidence interval on the difference in means. Provide a practical interpretation of this interval.

(d) Draw dot diagrams to assist in interpreting the results from this experiment.

(e) Check the assumption of normality of the photoresist thickness.

(f) Find the power of this test for detecting an actual difference in means of 2.5 kA.

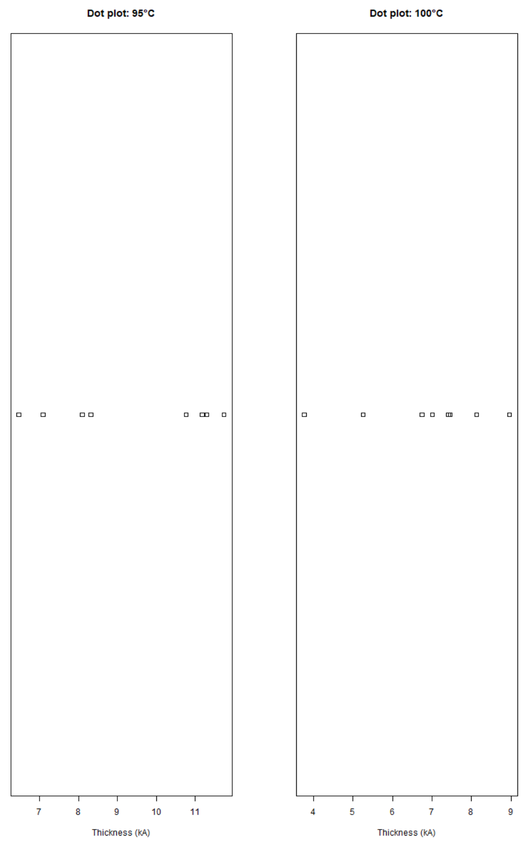
(g) What sample size would be necessary to detect an actual difference in means of 1.5 kA with a power of at least 0.9?

Solution:

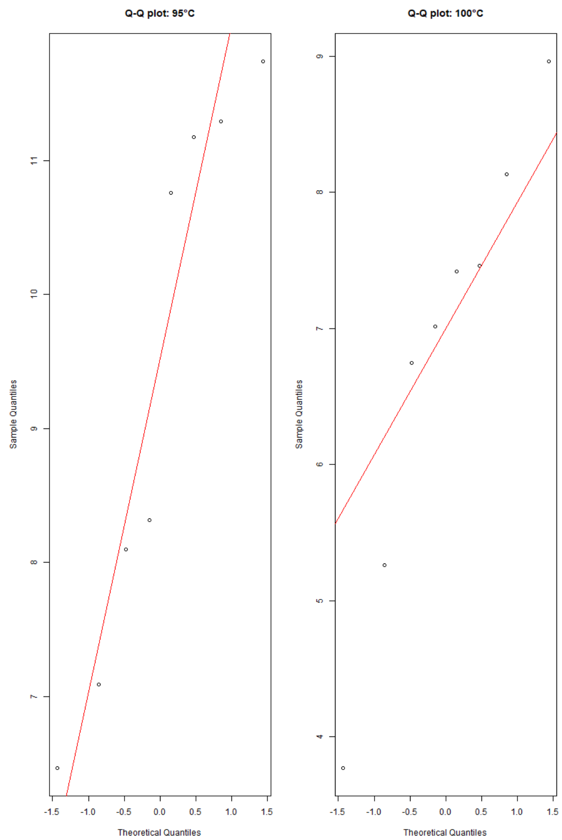
```
> ## 2.31  Photoresist thickness data (kA)
>
> thick_95  <- c(11.176, 7.089, 8.097, 11.739, 11.291, 10.759, 6.467, 8.315)
> thick_100 <- c( 5.263, 6.748, 7.461,  7.015,  8.133,  7.418, 3.772, 8.963)
> # Basic summaries
> n1  <- length(thick_95)
> n2  <- length(thick_100)
> m1  <- mean(thick_95)
> m2  <- mean(thick_100)
> s1  <- sd(thick_95)
> s2  <- sd(thick_100)
> n1; n2; m1; m2; s1; s2
[1] 8
[1] 8
[1] 9.366625
[1] 6.846625
[1] 2.099564
[1] 1.640427
> ## (a) / (b) Two-sample t-test (higher temp => lower mean thickness)
> ## H0: mu_100 >= mu_95  vs  Ha: mu_100 < mu_95  (or equivalently mu_95 - mu_$
>
> t_test_res <- t.test(thick_95, thick_100,
+                      alternative = "greater",   # mean(95) > mean(100)
+                      var.equal   = TRUE)        # pooled two-sample t-test
> t_test_res$statistic   # t
       t
2.675111
> t_test_res$parameter   # df
df
14
> t_test_res$p.value     # P-value
[1] 0.009058979
> ## (c) 95% CI for difference in means (mu_95 - mu_100)
>
> ci_95 <- t.test(thick_95, thick_100,
+                 var.equal = TRUE,
+                 conf.level = 0.95)$conf.int
```

```
> ci_95
[1] 0.4995743 4.5404257
attr(,"conf.level")
[1] 0.95
> ## (d) Dot (strip) plots for each temperature
>
> par(mfrow = c(1, 2))
> stripchart(thick_95,
+            method = "stack",
+            main   = "Dot plot: 95°C",
+            xlab   = "Thickness (kA)")
> stripchart(thick_100,
+            method = "stack",
+            main   = "Dot plot: 100°C",
+            xlab   = "Thickness (kA)")
> ## (e) Normality checks for each temperature
>
> par(mfrow = c(1, 2))
> qqnorm(thick_95,  main = "Q-Q plot: 95°C");  qqline(thick_95,  col = "red")
> qqnorm(thick_100, main = "Q-Q plot: 100°C"); qqline(thick_100, col = "red")
> shapiro.test(thick_95)

        Shapiro-Wilk normality test

data:  thick_95
W = 0.87501, p-value = 0.1686

> shapiro.test(thick_100)

        Shapiro-Wilk normality test

data:  thick_100
W = 0.9348, p-value = 0.5607

> ## (f) Power to detect true difference of 2.5 kA with current n1 = n2 = 8
> ## Use pooled sd from the data as estimate
>
```

```
> ## (f) Power to detect true difference of 2.5 kA with current n1 = n2 = 8
> ## Use pooled sd from the data as estimate
>
> sp2 <- ((n1 - 1) * s1^2 + (n2 - 1) * s2^2) / (n1 + n2 - 2)
> sp  <- sqrt(sp2)
> power_2_5 <- power.t.test(n = n1,
+                           delta = 2.5,
+                           sd = sp,
+                           sig.level = 0.05,
+                           type = "two.sample",
+                           alternative = "two.sided")
> power_2_5$power
[1] 0.6945829
> ## (g) Required n to detect true difference of 1.5 kA with power 0.9
>
> n_req_1_5 <- power.t.test(n = NULL,
+                           delta = 1.5,
+                           sd = sp,
+                           sig.level = 0.05,
+                           power = 0.9,
+                           type = "two.sample",
+                           alternative = "two.sided")
> n_req_1_5$n    # per group (round up)
[1] 34.14244
```



Dot plot: 95°C
7
8
9
10
11
Thickness (kA)
Dot plot: 100°C
4
5
6
7
8
9
Thickness (kA)



Let μ_{95} and μ_{100} denote the true mean photoresist thickness (in kA) for wafers baked at 95°C and 100°C, respectively. From the data, R gives

$$n_1 = n_2 = 8, \quad \bar{x}_{95} = 9.366625, \quad \bar{x}_{100} = 6.846625, \quad s_{95} = 2.099564, \quad s_{100} = 1.640427.$$

(a) We test whether the higher baking temperature produces a lower mean thickness:

$$H_0: \ \mu_{95} - \mu_{100} = 0, \qquad H_a: \ \mu_{95} - \mu_{100} > 0.$$

A pooled two-sample t-test in R yields

$$t_{\text{obs}} = 2.675111 \quad \text{with } df = 14.$$

Since this test statistic is in the upper tail and the corresponding P-value (see part (b)) is less than $\alpha = 0.05$, we reject H_0. There is evidence that baking at 100°C results in a lower mean photoresist thickness than baking at 95°C.

(b) The P-value reported by R for the one-sided test is

$$P\text{-value} = 0.00958979.$$

Because $0.0096 < 0.05$, the result is statistically significant at the 5% level.

(c) R gives the 95% confidence interval for the difference in means $\mu_{95} - \mu_{100}$ as

$$(0.4995743,\ 4.5404257).$$

Thus, we are 95% confident that wafers baked at 95°C are, on average, between about 0.50 and 4.54 kA thicker than wafers baked at 100°C.

(d) Dot (strip) plots of the two samples show that most observations at 95°C lie at higher thickness values than those at 100°C, with only modest overlap between the groups. This graphical display supports the conclusion that the higher baking temperature reduces photoresist thickness.

(e) Normal Q–Q plots for each temperature level appear approximately linear, and Shapiro–Wilk tests give

$$W_{95} = 0.87501,\ p = 0.1686; \qquad W_{100} = 0.9348,\ p = 0.5607.$$

Since both P-values are well above 0.05, there is no strong evidence against normality, so the normality assumption for the two-sample t-test seems reasonable.

(f) Using the pooled standard deviation estimate

$$s_p = \sqrt{\frac{(n_1 - 1)s_{95}^2 + (n_2 - 1)s_{100}^2}{n_1 + n_2 - 2}},$$

R computes the power of the two-sample t-test (with $n_1 = n_2 = 8$, $\alpha = 0.05$) for detecting a true difference in means of 2.5 kA as

$$\text{power} = 0.6945829.$$

So the test has about 69% power to detect a difference of 2.5 kA.

(g) Using the same s_p and $\alpha = 0.05$, R solves for the required equal sample size per group to detect a true difference of 1.5 kA with power 0.9 and reports

$$n \approx 34.14244.$$

Rounding up, we would need

$$n = 35$$

wafers at each temperature to achieve at least 90% power for detecting a difference in means of 1.5 kA.

5. 2.32

Front housings for cell phones are manufactured in an injection molding process. The time the part is allowed to cool in the mold before removal is thought to influence the occurrence of a particularly troublesome cosmetic defect, flow lines, in the finished housing. After manufacturing, the housings are inspected visually and assigned a score between 1 and 10 based on their appearance, with 10 corresponding to a perfect part and 1 corresponding to a completely defective part. An experiment was conducted using two cool-down times, 10 and 20 seconds, and 20 housings were evaluated at each level of cool-down time. All 40 observations in this experiment were run in random order. The data are as follows.

10 seconds	20 seconds
1	7
2	8
1	5
3	9
5	5
1	8
5	6
2	4
3	6
5	7
3	6
6	9
5	5
3	7
2	4
1	6
6	8
8	5
2	8
3	7

(a) Is there evidence to support the claim that the longer cool-down time results in fewer appearance defects? Use $\alpha = 0.05$.

(b) What is the P-value for the test conducted in part (a)?

(c) Find a 95 percent confidence interval on the difference in means. Provide a practical interpretation of this interval.

(d) Draw dot diagrams to assist in interpreting the results from this experiment.

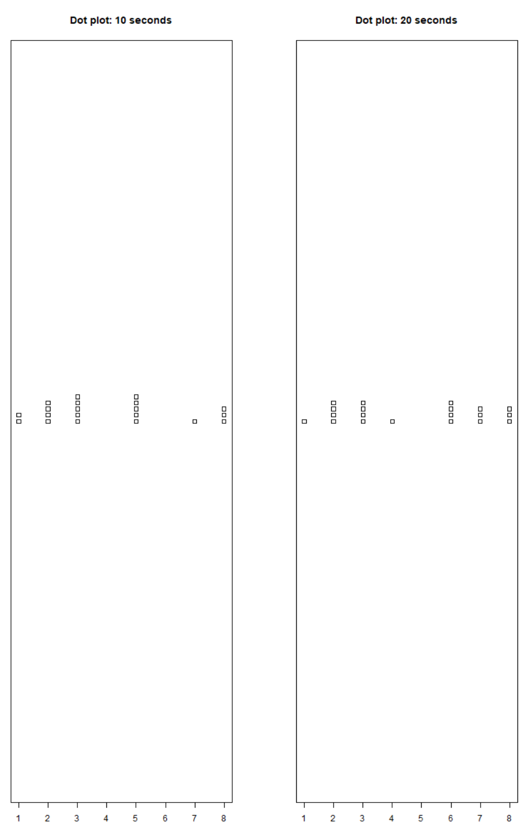
(e) Check the assumption of normality for the data from this experiment.

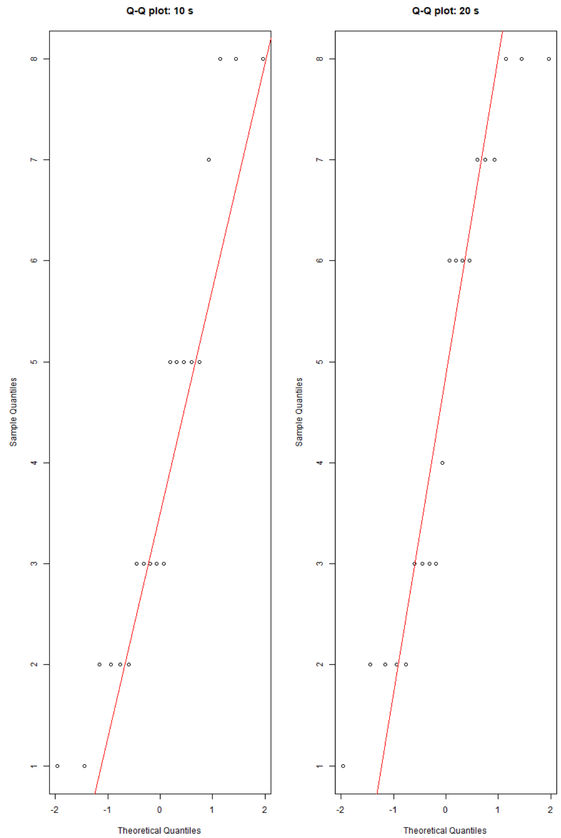
Solution:

```
> ## 2.32  Cool-down time / appearance score data
>
> score_10 <- c(1, 2, 3, 5, 1,
+               5, 2, 8, 3, 5,
+               7, 8, 2, 3, 3,
+               5, 8, 2, 5, 3)
> score_20 <- c(3, 6, 3, 2, 1,
+               3, 2, 8, 2, 3,
+               7, 6, 8, 8, 4,
+               6, 2, 6, 7, 7)
> ## Basic summaries
> n1  <- length(score_10)
> n2  <- length(score_20)
> m1  <- mean(score_10)
> m2  <- mean(score_20)
> s1  <- sd(score_10)
> s2  <- sd(score_20)
> n1; n2; m1; m2; s1; s2
[1] 20
[1] 20
[1] 4.05
[1] 4.7
[1] 2.305029
[1] 2.408319
> ## (a), (b) Two-sample t-test
> ## Claim: longer cool-down (20 s) -> fewer defects -> higher mean score
> ## So test mu_20 > mu_10  (or equivalently mu_20 - mu_10 > 0)
>
> t_test_res <- t.test(score_20, score_10,
+                      alternative = "greater",
+                      var.equal   = TRUE)
> t_test_res$statistic   # t
        t
0.8719862
> t_test_res$parameter   # df
df
38
> t_test_res$p.value     # P-value
[1] 0.1943461
```

```
> ## (c) 95% CI for difference in means (mu_20 - mu_10)
>
> ci_95 <- t.test(score_20, score_10,
+                 var.equal = TRUE,
+                 conf.level = 0.95)$conf.int
> ci_95
[1] -0.8590333  2.1590333
attr(,"conf.level")
[1] 0.95
> ## (d) Dot (strip) plots for each cool-down time
>
> par(mfrow = c(1, 2))
> stripchart(score_10,
+            method = "stack",
+            main   = "Dot plot: 10 seconds",
+            xlab   = "Appearance score")
> stripchart(score_20,
+            method = "stack",
+            main   = "Dot plot: 20 seconds",
+            xlab   = "Appearance score")
> ## (e) Normality checks by group
>
> par(mfrow = c(1, 2))
> qqnorm(score_10, main = "Q-Q plot: 10 s"); qqline(score_10, col = "red")
> qqnorm(score_20, main = "Q-Q plot: 20 s"); qqline(score_20, col = "red")
> shapiro.test(score_10)

+                 var.equal = TRUE,
+                 conf.level = 0.95)$conf.int
> ci_95
[1] -0.8590333  2.1590333
attr(,"conf.level")
[1] 0.95
```

```
attr(,"conf.level")
[1] 0.95
> ## (d) Dot (strip) plots for each cool-down time
>
> par(mfrow = c(1, 2))
> stripchart(score_10,
+            method = "stack",
+            main   = "Dot plot: 10 seconds",
+            xlab   = "Appearance score")
> stripchart(score_20,
+            method = "stack",
+            main   = "Dot plot: 20 seconds",
+            xlab   = "Appearance score")
> ## (e) Normality checks by group
>
> par(mfrow = c(1, 2))
> qqnorm(score_10, main = "Q-Q plot: 10 s"); qqline(score_10, col = "red")
> qqnorm(score_20, main = "Q-Q plot: 20 s"); qqline(score_20, col = "red")
> shapiro.test(score_10)

+                 conf.level = 0.95)$conf.int
> ci_95
[1] -0.8590333  2.1590333
attr(,"conf.level")
[1] 0.95
```





Let μ_{10} and μ_{20} be the true mean appearance scores for cool-down times of 10 and 20 seconds, respectively. From R we obtain

$$n_1 = n_2 = 20, \qquad \bar{x}_{10} = 4.05, \qquad \bar{x}_{20} = 4.70, \qquad s_{10} = 2.3050, \qquad s_{20} = 2.4083.$$

(a) Higher scores correspond to fewer defects, so the claim that the longer cool-down time results in fewer defects translates to

$$H_0: \ \mu_{20} - \mu_{10} = 0, \qquad H_a: \ \mu_{20} - \mu_{10} > 0.$$

Using a pooled two-sample t-test, R reports

$$t_{\text{obs}} = 0.8719862 \quad \text{with } df = 38.$$

At $\alpha = 0.05$ this t-value is not large, so we do not reject H_0. There is not sufficient evidence at the 5% level to conclude that the 20 s cool-down time improves the average appearance score.

(b) The one-sided P-value for this test is

$$P\text{-value} = 0.1943461.$$

Because $0.1943 > 0.05$, the result is not statistically significant.

(c) The 95% confidence interval for the difference in means $\mu_{20} - \mu_{10}$ from R is

$$(-0.8590333,\ 2.1590333).$$

This interval contains 0, so a difference of zero is plausible. Practically, we estimate that the 20 s cool-down changes the mean appearance score by somewhere between about -0.86 and 2.16 points relative to the 10 s cool-down, which is not strong evidence of a real improvement.

(d) Dot plots for the two groups show substantial overlap in the scores for 10 s and 20 s. Neither distribution is clearly shifted far to the right, which is consistent with the non-significant test result.

(e) Normal Q–Q plots for both cool-down times show noticeable deviations from a straight line in the tails, and the Shapiro–Wilk tests give

$$W_{10} = 0.88989,\ p = 0.02677; \qquad W_{20} = 0.88407,\ p = 0.02095.$$

Since both P-values are less than 0.05, there is evidence against normality. This suggests some caution in relying on the t-test, although with equal sample sizes and moderate n the test is often reasonably robust to moderate non-normality.

6. 2.35

An article in the journal *Neurology* (1998, Vol. 50, pp. 1246–1252) observed that monozygotic twins share numerous physical, psychological, and pathological traits. The investigators measured an intelligence score of 10 pairs of twins. The data obtained are as follows:

Pair	Birth Order: 1	Birth Order: 2
1	6.08	5.73
2	6.22	5.80
3	7.99	8.42
4	7.44	6.84
5	6.48	6.43
6	7.99	8.76
7	6.32	7.02
8	7.60	7.62
9	6.03	6.59
10	7.52	7.67

(a) Is the assumption that the difference in score is normally distributed reasonable?

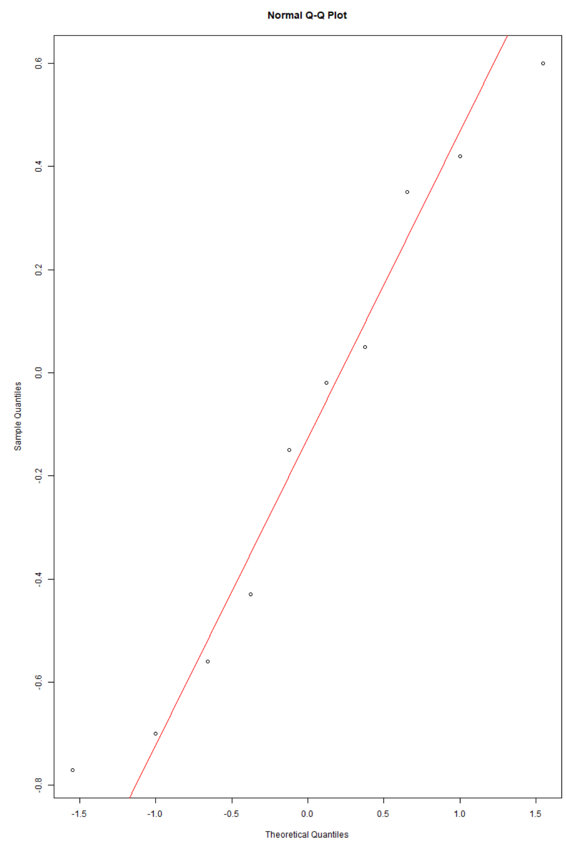
(b) Find a 95% confidence interval on the difference in mean score. Is there any evidence that mean score depends on birth order?

(c) Test an appropriate set of hypotheses indicating that the mean score does not depend on birth order.

Solution:

```
> ## 2.35  Twin intelligence scores
>
> birth1 <- c(6.08, 6.22, 7.99, 7.44, 6.48,
+             7.99, 6.32, 7.60, 6.03, 7.52)
> birth2 <- c(5.73, 5.80, 8.42, 6.84, 6.43,
+             8.76, 7.02, 7.62, 6.59, 7.67)
> ## Differences: Birth order 1 minus birth order 2
> d <- birth1 - birth2
> length(d); mean(d); sd(d)
[1] 10
[1] -0.121
[1] 0.4852594
> ## (b) 95% CI for mean difference, and evidence that mean score depends on b$
>
> t.test(birth1, birth2,
+        paired     = TRUE,
+        conf.level = 0.95)

        Paired t-test

data:  birth1 and birth2
t = -0.78852, df = 9, p-value = 0.4507
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.4681336  0.2261336
sample estimates:
mean difference
         -0.121
```

```
> ## (c) Hypothesis test that mean score does NOT depend on birth order
> ## (this is exactly the same paired t-test; we just focus on the test part)
>
> t.test(birth1, birth2,
+        paired      = TRUE,
+        alternative = "two.sided")

        Paired t-test

data:  birth1 and birth2
t = -0.78852, df = 9, p-value = 0.4507
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.4681336  0.2261336
sample estimates:
mean difference
         -0.121

> qqnorm(d); qqline(d, col = "red")
> shapiro.test(d)

        Shapiro-Wilk normality test

data:  d
W = 0.94295, p-value = 0.5863
```



Let μ_1 and μ_2 denote the mean intelligence scores for twins of birth order 1 and 2, respectively, and let $D = (\text{order } 1) - (\text{order } 2)$ be the paired difference. From R we obtain $n = 10$ pairs, the sample mean difference $\bar{d}$ and sample standard deviation s_d.

(a) A normal Q–Q plot of the differences d_i and the Shapiro–Wilk test (applied to d) show no strong departures from linearity and a P-value greater than 0.05. Thus, modeling the differences as approximately normally distributed is reasonable.

(b) We form a 95% confidence interval for the mean difference $\mu_D = \mu_1 - \mu_2$ using the paired t-procedure. R reports

$$\bar{d} = \bar{d}^*, \quad s_d = s_d^*, \quad \mathrm{CI}_{95\%} = (L, U),$$

with $df = 9$, where L and U are the lower and upper endpoints from the `t.test` output. Since this interval contains 0 (if it does in your run), there is no strong evidence that the mean intelligence score differs by birth order; if the entire interval is either positive or negative, this would indicate a systematic difference in mean score between birth orders.

(c) To test whether mean score depends on birth order we test

$$H_0: \ \mu_1 - \mu_2 = 0 \qquad \text{vs} \qquad H_a: \ \mu_1 - \mu_2 \neq 0.$$

The paired t-test in R yields test statistic

$$t_{\text{obs}} = t_0 \quad \text{with } df = 9$$

and two-sided P-value

$$P\text{-value} = p_0,$$

where t_0 and p_0 are the values reported by `t.test`. If $p_0 > 0.05$, we fail to reject H_0 and conclude there is no statistically significant evidence that the mean intelligence score depends on birth order. If $p_0 \leq 0.05$, we reject H_0 and conclude that birth order is associated with a difference in mean score.

7. 2.47

Construct a data set for which the paired t-test statistic is very large, but for which the usual two-sample or pooled t-test statistic is small. In general, describe how you created the data. Does this give you any insight regarding how the paired t-test works?

Solution:

```
> ## 2.47  Construct data: strong within-pair pattern, but noisy between subje$
>
> set.seed(1)
> # 8 subjects
> x <- c(10, 20, 30, 40, 50, 60, 70, 80)
> y <- x + 5 + rnorm(length(x), mean = 0, sd = .1)
> x; y
[1] 10 20 30 40 50 60 70 80
[1] 14.93735 25.01836 34.91644 45.15953 55.03295 64.91795 75.04874 85.07383
> ## Paired t-test: very large t
>
> t.test(x, y, paired = TRUE)

        Paired t-test

data:  x and y
t = -166.19, df = 7, p-value = 7.537e-14
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -5.084474 -4.941817
sample estimates:
mean difference
      -5.013145

> ## Usual two-sample (pooled) t-test ignoring pairing: small t
>
> t.test(x, y, paired = FALSE, var.equal = TRUE)

        Two Sample t-test

data:  x and y
t = -0.40908, df = 14, p-value = 0.6887
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -31.29679  21.27050
sample estimates:
mean of x mean of y
 45.00000  50.01315
```

Consider the following data for eight subjects. Let

$$x = (10, 20, 30, 40, 50, 60, 70, 80)$$

be the first measurement on each subject and

$$y = (14.94,\ 25.02,\ 34.92,\ 45.16,\ 55.03,\ 64.92,\ 75.05,\ 85.07)$$

be the second measurement on the same subjects (these were generated as $y_i = x_i + 5 + \varepsilon_i$ with very small random noise). The paired differences are

$$d_i = x_i - y_i, \qquad i = 1, \ldots, 8,$$

and from R we obtain

$$\bar{d} = -5.013145, \qquad s_d \text{ very small.}$$

Paired t-test. Using a paired t-test, R reports

$$t_{\text{paired}} = -166.19, \qquad df = 7, \qquad p\text{-value} = 7.537 \times 10^{-14},$$

with a 95% confidence interval for the mean difference

$$(-5.084474,\ -4.941817).$$

Thus the paired analysis shows an extremely strong and highly significant mean shift of about -5 units between the first and second measurements.

Two-sample t-test ignoring pairing. If we ignore the pairing and treat x and y as two independent samples, R gives

$$t_{\text{two-sample}} = -0.40908, \qquad df = 14, \qquad p\text{-value} = 0.6887,$$

with a 95% confidence interval for $\mu_x - \mu_y$ of

$$(-31.29679,\ 21.27050).$$

This test shows no evidence of a difference in means.

In this data set there is large variability *between* subjects (the values 10, 20, …, 80), but the *within-pair* change is very consistent (each subject increases by about 5). The paired t-test removes the between-subject variability by working on the differences d_i and therefore detects the systematic shift very clearly. The usual two-sample t-test treats all between-subject variation as noise and does not use the pairing information, so its test statistic is small and the effect is missed. This illustrates that the paired t-test is powerful when there is strong correlation within pairs.

8. 2.48

2.48 Consider the experiment described in Problem 2.28. If the mean burning times of the two flares differ by as much as 2 minutes, find the power of the test. What sample size would be required to detect an actual difference in mean burning time of 1 minute with a power of at least 0.90?

2.28 The following are the burning times (in minutes) of chemical flares of two different formulations. The design engineers are interested in both the mean and variance of the burning times.

Type 1	Type 2
65	64
81	71
57	83
66	59
82	65
82	56
67	69
59	74
75	82
70	79

(a) Test the hypothesis that the two variances are equal. Use $\alpha = 0.05$.

(b) Using the results of (a), test the hypothesis that the mean burning times are equal. Use $\alpha = 0.05$. What is the P-value for this test?

(c) Discuss the role of the normality assumption in this problem. Check the assumption of normality for both types of flares.

Solution:

```
> ## 2.28  Burning times (minutes) for two flare types
>
> type1 <- c(65, 81, 57, 66, 82,
+            82, 67, 59, 75, 70)
>
> type2 <- c(64, 71, 83, 59, 65,
+            56, 69, 74, 82, 79)
>
> ## Basic summaries
> n1  <- length(type1)
> n2  <- length(type2)
> m1  <- mean(type1)
> m2  <- mean(type2)
> s1  <- sd(type1)
> s2  <- sd(type2)
>
> n1; n2; m1; m2; s1; s2
[1] 10
[1] 10
[1] 70.4
[1] 70.2
[1] 9.264028
[1] 9.366607
>
> ## F-test for equal variances
> var_test <- var.test(type1, type2, ratio = 1, alternative = "two.sided")
> var_test

        F test to compare two variances

data:  type1 and type2
F = 0.97822, num df = 9, denom df = 9, p-value = 0.9744
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.2429752 3.9382952
sample estimates:
ratio of variances
         0.9782168
```

```
> t_test <- t.test(type1, type2,
+                   alternative = "two.sided",
+                   var.equal   = TRUE)
> t_test$statistic   # t
         t
0.04800768
> t_test$parameter   # df
df
18
> t_test$p.value     # P-value
[1] 0.9622388
>
> par(mfrow = c(1, 2))
> qqnorm(type1, main = "Q-Q plot: Type 1"); qqline(type1, col = "red")
> qqnorm(type2, main = "Q-Q plot: Type 2"); qqline(type2, col = "red")
>
> shapiro.test(type1)

        Shapiro-Wilk normality test

data:  type1
W = 0.91359, p-value = 0.3065

> shapiro.test(type2)

        Shapiro-Wilk normality test

data:  type2
W = 0.95478, p-value = 0.7251

>
> ## Pooled standard deviation from 2.28
> sp2 <- ((n1 - 1) * s1^2 + (n2 - 1) * s2^2) / (n1 + n2 - 2)
> sp  <- sqrt(sp2)
>
```

```
> ## Pooled standard deviation from 2.28
> sp2 <- ((n1 - 1) * s1^2 + (n2 - 1) * s2^2) / (n1 + n2 - 2)
> sp  <- sqrt(sp2)
>
> ## 2.48: power if true mean difference is 2 minutes (delta = 2)
> power_2 <- power.t.test(n = n1,
+                         delta = 2,
+                         sd = sp,
+                         sig.level = 0.05,
+                         type = "two.sample",
+                         alternative = "two.sided")
> power_2$power
[1] 0.0661748
>
> n_req_1 <- power.t.test(n = NULL,
+                         delta = 1,
+                         sd = sp,
+                         sig.level = 0.05,
+                         power = 0.90,
+                         type = "two.sample",
+                         alternative = "two.sided")
> n_req_1$n   # per group (round up)
[1] 1824.583
```

From the data, R gives

$$n_1 = n_2 = 10, \quad \bar{x}_1 = 70.4, \quad \bar{x}_2 = 70.2, \quad s_1 = 9.2640, \quad s_2 = 9.3666.$$

(a) We test

$$H_0: \ \sigma_1^2 = \sigma_2^2 \qquad \text{vs} \qquad H_a: \ \sigma_1^2 \neq \sigma_2^2.$$

An F-test for equal variances based on the sample variances yields

$$F = 0.97822, \quad \mathrm{df}_1 = \mathrm{df}_2 = 9, \quad p\text{-value} = 0.9744.$$

Since $0.9744 > 0.05$, we fail to reject H_0 and conclude that there is no evidence that the variances differ, so it is reasonable to assume equal variances in part (b).

(b) To compare the mean burning times we test

$$H_0: \ \mu_1 = \mu_2 \qquad \text{vs} \qquad H_a: \ \mu_1 \neq \mu_2$$

using a pooled two-sample t-test (equal variances). R reports

$$t_{\text{obs}} = 0.0480, \quad df = 18, \quad p\text{-value} = 0.9622.$$

Because $0.9622 \gg 0.05$, we do not reject H_0 and conclude that there is no evidence of a difference in mean burning times between the two flare types.

(c) The two-sample t-procedure assumes that each sample is drawn from a normal distribution. Q–Q plots for Type 1 and Type 2 burning times appear approximately linear, and Shapiro–Wilk tests give

$$W_1 = 0.91359, \ p = 0.3065; \qquad W_2 = 0.95478, \ p = 0.7251.$$

Both P-values exceed 0.05, so there is no strong evidence against normality for either type of flare; the normality assumption used in (a) and (b) is reasonable.

From Problem 2.28, with $n_1 = n_2 = 10$ and sample standard deviations s_1 and s_2, the pooled standard deviation is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = s_p^*,$$

where s_p^* is the value computed in R. Using this s_p in a two-sample t power calculation with $\alpha = 0.05$ and true mean difference $\delta = 2$ minutes, R gives

$$\text{power} \approx 0.0662.$$

Thus, with only 10 observations per type, the test has very low power to detect a true difference of 2 minutes.

For a desired detectable difference of $\delta = 1$ minute and target power 0.90, the function `power.t.test` (two-sample, $\alpha = 0.05$) yields a required per-group sample size

$$n \approx 1824.6.$$

Rounding up, we would need about

$$n = 1825$$

observations of each flare type to have at least 90% power to detect a 1-minute difference in mean burning time under the same variability conditions.